

Abstract

Transition metal dichalcogenides (TMDs) have emerged as leading candidates for next-generation electronic, spintronic, and catalytic applications, owing to their stable two-dimensional hexagonal structure and highly tunable electronic and optical properties. Among TMDs, tungsten disulfide (WS_2) is of particular interest due to its sizeable bandgap and strong spin-orbit coupling. Point defects, especially chalcogen vacancies, occur naturally in 2D materials and can also be introduced deliberately through α -particle or electron-beam irradiation and thermal annealing, offering a practical route to tailor material properties and providing insight into the effects of substitutional doping.

In this work, we perform first-principles density functional theory calculations on a $4 \times 4 \times 1$ monolayer WS_2 supercell to systematically investigate the influence of sulfur vacancies on the electronic structure. The pristine monolayer yields a bandgap of 1.875 eV, in good agreement with previously reported values. Sulfur vacancy concentrations of 3%, 6%, 9%, and 12% are then introduced, and the resulting modifications to the band structure and density of states are analyzed. Our results elucidate the concentration-dependent evolution of vacancy-induced states within the gap, providing guidance for defect-engineered design of WS_2 -based devices.

Introduction

The rapid growth of the semiconductor industry has increased demand for materials with tunable electronic, optical, and magnetic properties. Conventional bulk semiconductors, especially silicon, are reaching limits in miniaturization, energy efficiency, and multifunctionality [1]. To meet next-generation electronic, optoelectronic, spintronic, and quantum device requirements, alternative materials are needed. In this context, two-dimensional (2D) semiconductors, particularly transition metal dichalcogenides (TMDs), are promising due to their reduced dimensionality and unique properties.

A key motivation for studying these materials is the occurrence of electronic phase transitions. In strongly correlated systems and low-dimensional materials, small changes in temperature, pressure, strain, or carrier concentration can induce semiconducting, half-metallic, or spin-gapless states, accompanied by significant variations in electrical, optical, spintronic, and magnetic properties. Controlling these transitions offers potential for advanced memory, sensing, and adaptive electronic devices.

Since graphene's discovery in 2004, 2D materials have drawn attention for their thickness-dependent electronic properties. TMDs (MX_2) show an indirect-to-direct bandgap transition at the monolayer limit, e.g., MoS_2 from ~ 1.29 eV (bulk) to ~ 1.8 eV (monolayer). WS_2 is notable for strong spin-orbit coupling, a direct ~ 2 eV bandgap, and superior optical/electronic properties, suitable for optoelectronic and nanoelectronic applications. Defect engineering tunes TMD behavior: chalcogen vacancies create localized states near the Fermi level, while oxygen doping passivates defects and stabilizes the structure. This work employs DFT to study pristine and defected WS_2 , focusing on sulfur vacancies and oxygen doping effects.

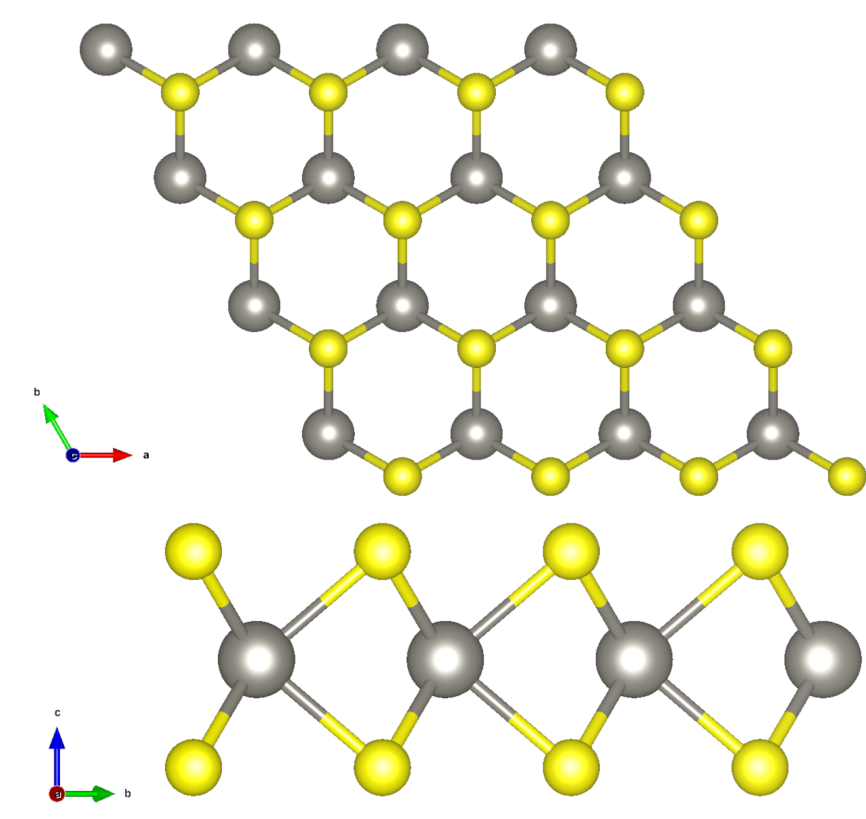


Figure 1. Top and side views of single-layer WS_2 supercell. Gray and yellow spheres represent tungsten (W) and sulfur (S) atoms.

Methodology

First-principles calculations were performed to study pristine, sulfur-vacancy, and oxygen-doped WS_2 monolayers. The $4 \times 4 \times 1$ pristine supercell contains 48 atoms (16 W and 32 S) with a layered structure, where a tungsten plane is sandwiched between two sulfur planes. A 20 Å vacuum was applied along the $\vec{z}$ direction to avoid interlayer interactions. The GGA in the PBE form was used for exchange-correlation, along with the PAW method. To account for the localized W d-electrons, DFT+U with $U = 2.87$ eV was applied. Sulfur vacancies (V_S) were introduced systematically at concentrations of 3%, 6%, 9%, and 12% by removing 1–4 S atoms (Fig. 1). A plane-wave cutoff of 520 eV was employed, and the Brillouin zone was sampled with a $1 \times 1 \times 1$ k-point mesh for structural optimization and $9 \times 9 \times 1$ for electronic structure calculations.

Structural relaxation was performed with fixed lattice parameters, allowing atomic positions to relax until forces on all atoms were below 0.01 eV/Å and total energy converged within 10^{-5} eV. Spin-polarized calculations were carried out to capture magnetic effects induced by sulfur vacancies (V_S) and oxygen doping (V_S^O). The electronic structure was analyzed using spin-resolved total and partial density of states to examine defect- or doping-induced polarization. Results were validated against available theoretical and experimental data.

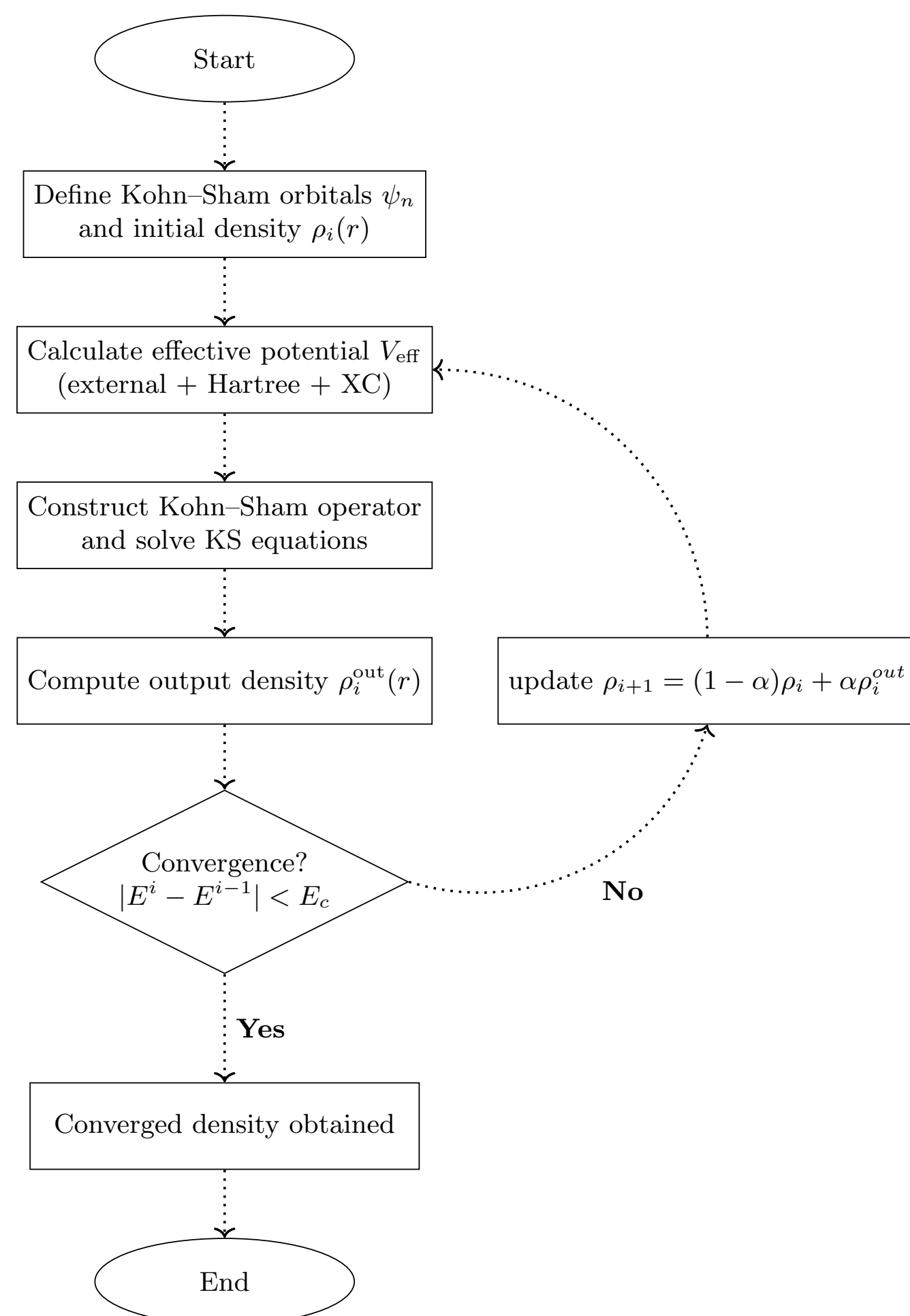


Figure 2. Schematic representation of the self-consistent field (SCF) loop used in density functional theory (DFT) calculations. The flowchart illustrates the iterative procedure for solving the Kohn-Sham equations.

Result & Discussion

Structural optimization of the WS_2 unit cell yields lattice parameters and bond lengths in good agreement with previously reported theoretical and experimental results. Analysis of the total and projected density of states (DOS and PDOS) confirms that pristine monolayer WS_2 exhibits semiconducting behavior with a band gap of approximately 1.86 eV and no spin polarization, indicating a non-magnetic ground state. The valence band is primarily dominated by S-p orbitals with hybridization from W-d states, while the conduction band is mainly composed of W-d orbitals.

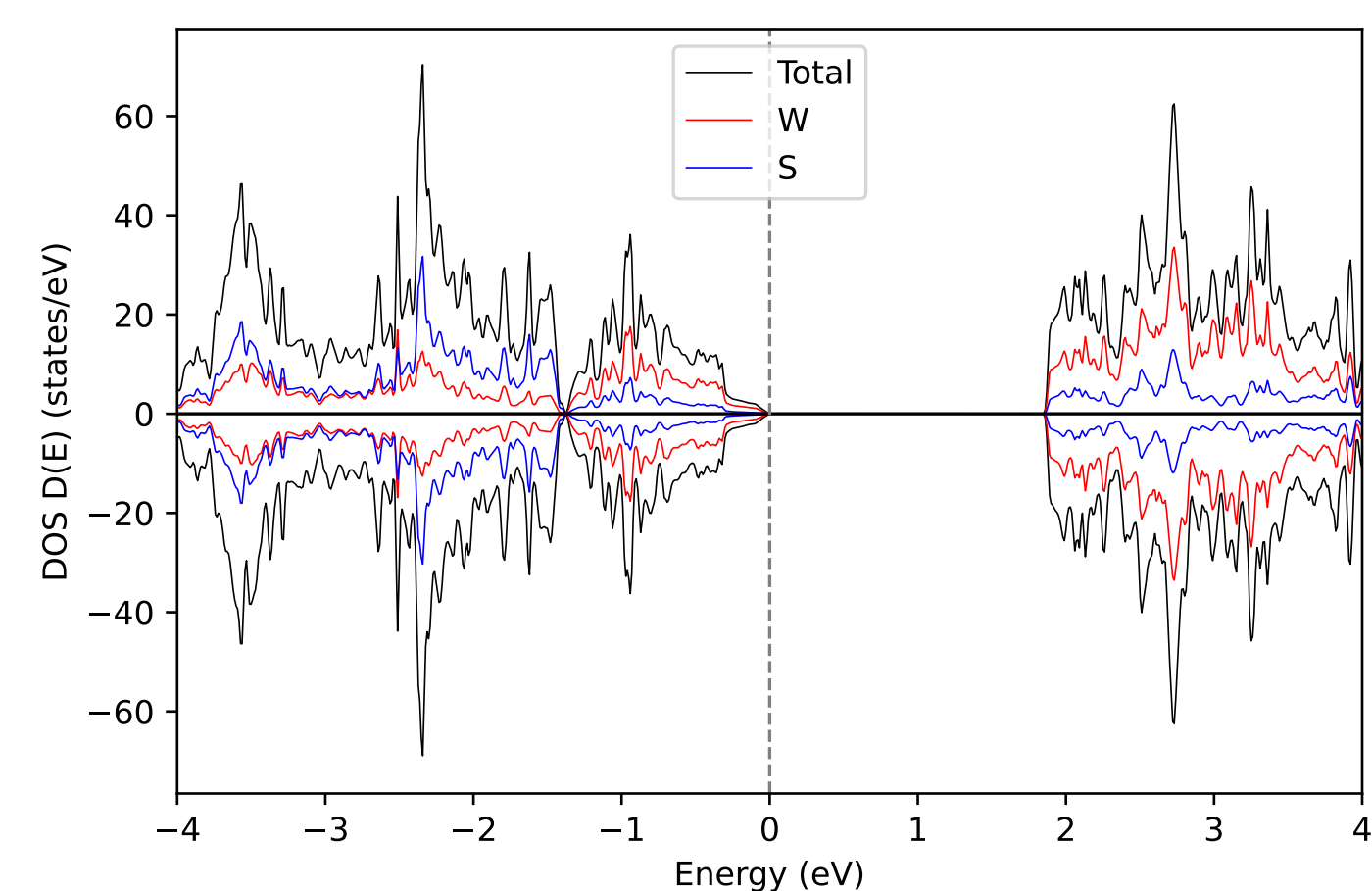


Figure 3. The total and partial density of states (DOS and PDOS) for the pristine WS_2 system are presented. In each plot, the dotted line denotes the Fermi level.

Introduction of sulfur vacancies leads to the emergence of defect-induced states near the Fermi level and a reduction in the band gap; however, no magnetism is observed due to the delocalized nature of these states. In contrast, oxygen substitution at sulfur vacancy sites effectively passivates these defect states, suppresses mid-gap states, and restores semiconducting behavior without inducing magnetism. Overall, oxygen doping stabilizes the electronic structure of WS_2 by eliminating defect states and maintaining a nearly non-magnetic ground state.

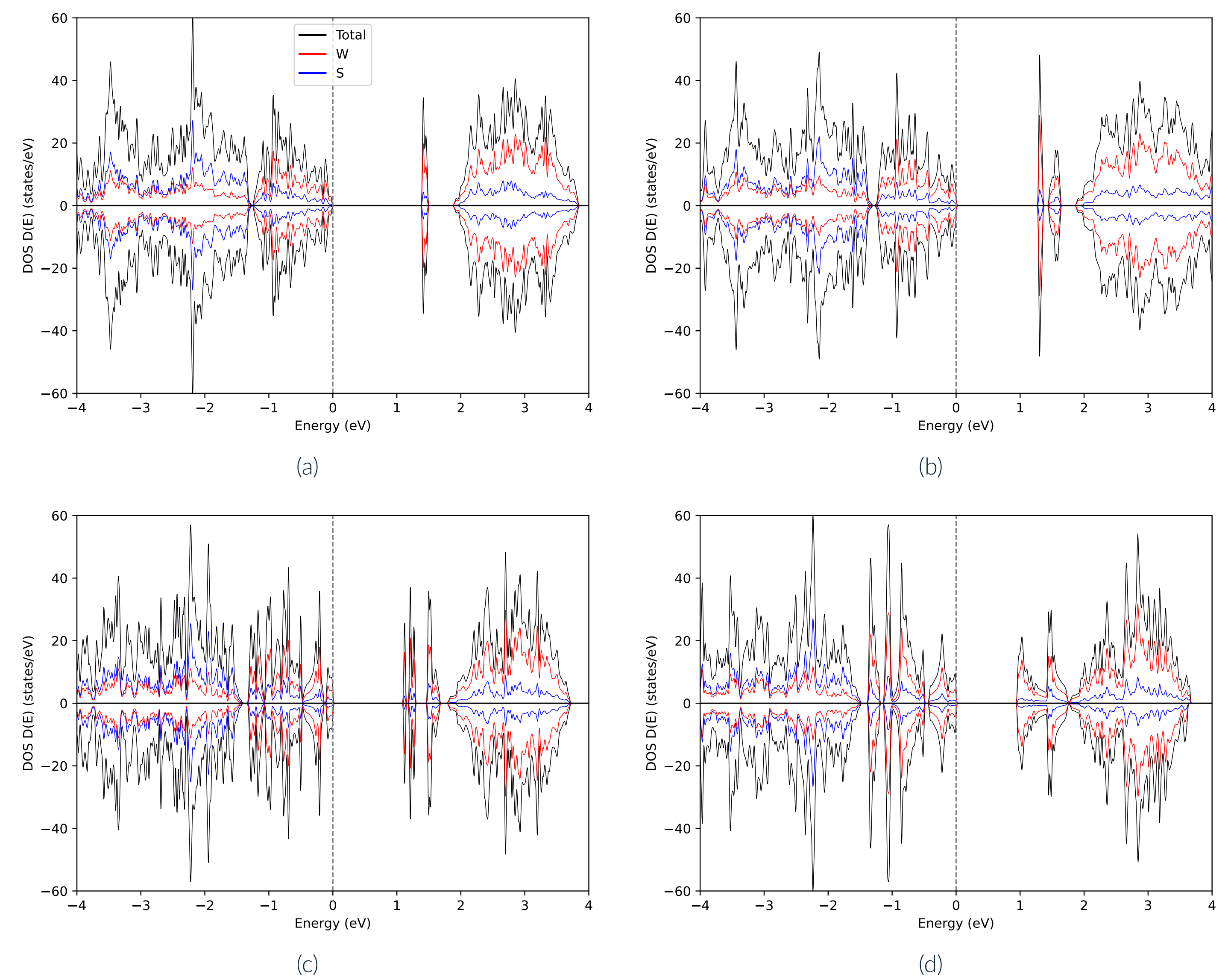


Figure 4. The total and Partial density of states (DOS and PDOS) plots are shown for vacancy-induced WS_2 systems with (a) 3%, (b) 6%, (c) 9%, and (d) 12% vacancy concentrations. In each plot, the dotted line indicates the Fermi level.

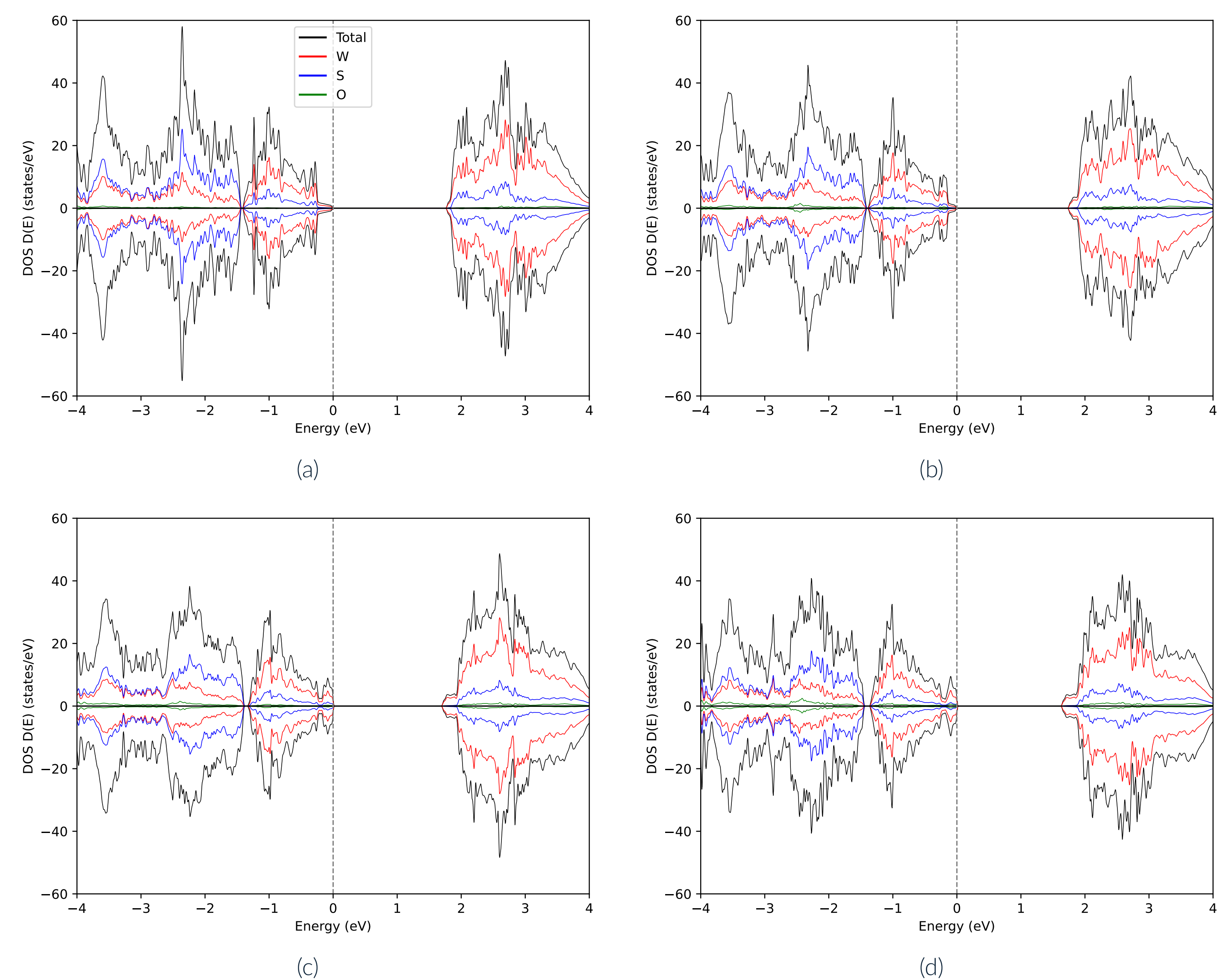


Figure 5. The total and partial density of states (DOS and PDOS) are shown for oxygen-doped WS_2 systems with doping concentrations of (a) 3%, (b) 6%, (c) 9%, and (d) 12%. In each plot, the dotted line marks the Fermi level.

Conclusion

The pristine monolayer is a direct-gap semiconductor (1.86 eV) with tungsten d-dominated band edges and no spin polarization. Sulfur vacancies create defect states near the Fermi level, reducing the bandgap without magnetism, while oxygen doping passivates dangling bonds, preserves tungsten d character, places oxygen p states in the valence region, and maintains a non-magnetic state. These results show that sulfur vacancies degrade, whereas oxygen doping stabilizes, the electronic structure, demonstrating tunable WS_2 via defect engineering.

Scope of Future work

As a continuation of this work, further investigations will focus on inducing magnetism in WS_2 through alternative defect engineering strategies. In particular, the creation of tungsten vacancies (V_W) and substitutional doping with transition metal atoms at tungsten sites will be explored. Unlike sulfur vacancies, which do not generate spin polarization, tungsten-site modifications are expected to introduce localized d-electron states that can give rise to finite magnetic moments.

References

- [1] Shuiyuan Wang, Xiaoxian Liu, and Peng Zhou. The road for 2d semiconductors in the silicon age. *Advanced Materials*, 34(48):2106886, 2022.